

JAI BHAGAT

<https://jkbhagatio.io>
[jkbhagatio\[at\]gmail\[dot\]com](mailto:jkbhagatio[at]gmail[dot]com)

SELECTED SKILLS

ML & AI

Mechanistic Interpretability
Deep & Reinforcement Learning
(incl. Multimodal Transformers,
LSTMs, CNNs, TD, DQNs, PPO, IQL)
Supervised & Unsupervised Learning
(incl. GLMs, SVMs, Forests,
HDBSCAN, OPTICS)
Distributed Training
(incl. DDP, FSDP over a full 72-GPU
cluster via Dstack, Skypilot)
Cloud Services
(incl. Wandb, AWS, GCP, HF)

Data Engineering

ETL Pipelines
Data Lakes
(Petabyte multimodal object stores)
Containerization & Workflow
Orchestration
(via Docker, K8s, Ray, Prefect)

Programming Languages

Python (incl. PyTorch, Jax),
Bonsai, C, Rust, CUDA

Mechatronics

Micro-controllers/computers
(incl. Arduino, R Pi)
Simple PID, KF Control Systems
Electrophysiology Acquisition
CAD & 3d Printing
Laser Cutting

Wet Lab

In-vivo Electrophysiology
Genotyping
Optogenetics
Stereotaxic surgeries
Histology

SELECTED AWARDS

Bogue Fellowship 2024
Fondation JFMLCT 2023
UCL AWPO 2022
SWC Public Engagement Fund 2022
SWC Ph. D. Scholarship 2021

EXTRA TRAINING

MARS Scholar - AI Alignment
ARENA scholar - AI Alignment
Machine Learning Summer School
Cajal Ephys Acquisition Course
Advanced Study Program - MIT

EDUCATION

Ph. D. Computational Neuroscience | University College London 2025
B.A. Neuroscience | Boston University 2015

SELECTED PROFESSIONAL EXPERIENCE

Founding Machine Learning Researcher, [Enigma](#) / [Metamorphic](#) 2025/09 – Present
Palo Alto, CA, USA
Bogue Fellow Researcher, [UCL - Anthropic](#) 2025/01 – 2025/03
San Francisco, CA, USA
Software Developer, [CortexLab](#) & [International Brain Lab](#) 2018/10 – 2020/08
University College London, London, UK
Technical Associate I/II, [Wilson Lab](#) 2016/06 – 2018/06
Massachusetts Institute of Technology, Cambridge, MA, USA

SELECTED PUBLICATIONS

F. D'Agostino*, A. Gilbert*, S. Keller*, ... **J. Bhagat**, et al. [Evaluating Neural Data Tokenizers: A Framework for Assessing Learned Representations of Spiking Activity](#). *NeurIPS 2026 (Oral)*.
F. D'Agostino, **J. Bhagat**, et al. SpikeDiT: Generative neural system identification at spike-train resolution. (*In Review*).
A. G. Pouget*, **J. Bhagat***, et al. Reinforcement learning models of ethological mouse behavior reveals static and dynamic social preferences and policies. (*In Review*).
J. Bhagat*, S. Molas-Medina*, G. Giglemiani, S. Heimersheim. [Compressed computation is not computation in superposition](#). *NeurIPS Mechanistic Interpretability Workshop 2025*.
J. Bhagat, A. G. Pouget, S. Molas-Medina. [A pipeline for interpretable neural latent discovery](#). *NeurIPS Data on the Brain & Mind Workshop 2025*.
D. Campagner*, **J. Bhagat***, G. Lopes*, et al. [Aeon: An open-source platform to study the neural basis of ethological behaviors over naturalistic timescales](#). *Biorxiv 2025 & In Press*.
International Brain Laboratory, K. Banga, J. Benson, **J. Bhagat**, et al. [Reproducibility of in-vivo electrophysiological measurements in mice](#). *eLife 2025*.
N. Steinmetz*, C. Aydin*, A. Lebedeva*, M. Okun*, M. Pachitariu*, **J. Bhagat**, et al. [Neuropixels 2.0: A high-density probe for stable, long-term brain recordings](#). *Science 2021*.
J. Bhagat*, M. J. Wells*, et al. [Rigbox: An open-source toolbox for probing neurons and behavior](#). *eNeuro 2020*.
J. Bhagat, et al. [LSTM neural networks for LFP event detection and classification in the rodent hippocampal-cortical network](#). *MIT BCS Symposium 2018*.

SELECTED OPEN-SOURCE PROJECTS

Neuronauts: An educational outreach program for teaching teenagers fundamentals in engineering, computer science, neuroscience, and artificial intelligence. (*Founder*)
nanoGPT: A minimal (nanomal?) Python repository containing code for building, training, and running nanoGPT. (*Sole creator, developer, maintainer*)
Wall-E-GPT: Python and Arduino code for a GPT-controlled, semi-autonomous rover robot running on a Raspberry Pi. (*Sole creator, developer, maintainer*)
aeon_mecha: [Project Aeon](#)'s main Python library for interfacing with acquired experiment data. (*Creator, developer, maintainer*)
aeon_experiments: [Project Aeon](#)'s main Bonsai and C# library for running behavioral neuroscience experiment workflows. (*Developer, maintainer*)
ibllib: The International Brain Laboratory's core shared Python libraries for data pipeline management and analysis. (*Developer, maintainer*)
Rigbox: A MATLAB and C based toolbox for running behavioral neuroscience experiments and managing data. (*Developer, maintainer*)